

CertLI: First Experiments

Can a compressed late-interaction index prove it returns the exact MaxSim top-10?

Complete code repository: github.com/RS-010806/certLI

What I ran. I implemented CertLI end to end and tested it on two ColBERT models (ColBERTv2 and AnswerAI-ColBERT-small) and two BEIR corpora (SciFact and NFCorpus). The pipeline has five parts: the corpus-mean shift (Lemma 1), low-rank-plus-sparse compression, interval bounds on MaxSim (Lemma 2), exact rescoring in the order of the bounds, and a calibrated variant using Learn-then-Test [3]. All results are measured against exact MaxSim over the whole corpus, so nothing depends on a candidate generator.

Findings.

1. **The math holds on real models.** Subtracting the corpus mean left every ranking unchanged, and the bounds were never violated in 50.3M (query, document) checks.
2. **Deterministic certificates are too costly at useful compression.** With the index compressed to about one third of its size, proving the exact top-10 still requires scoring 79% to 95% of the corpus for ColBERTv2 and all of it for AnswerAI-small (Figure 1, grey).
3. **The reason is clear in the data.** The coding error points in a nearly random direction relative to the query, so it is typically 5% of its worst-case bound, while the 10th and 11th exact scores are separated by only 0.08 (ColBERTv2) or less than 0.01 (AnswerAI).
4. **Calibrated certificates work.** Shrinking the same intervals by a factor chosen with Learn-then-Test (at most 5% of queries with a wrong top-10, with 90% confidence) cuts the cost to 1.4% of the corpus on ColBERTv2/SciFact, with 0.7% of held-out queries wrong (Figure 1, blue). A fixed rerank depth calibrated the same way is equally cheap, so the gain comes from calibration rather than from adapting to each query; a perfect stopping rule would need 3 to 7 times fewer documents, which is the room left for improvement.
5. **A finding relevant to EigenLI [2].** AnswerAI-small tokens share one dominant direction. Removing it is exactly safe for MaxSim but not for spectral scores: removing it from documents only drops an EigenLI-style score from 68.0 to 1.6 nDCG@10.

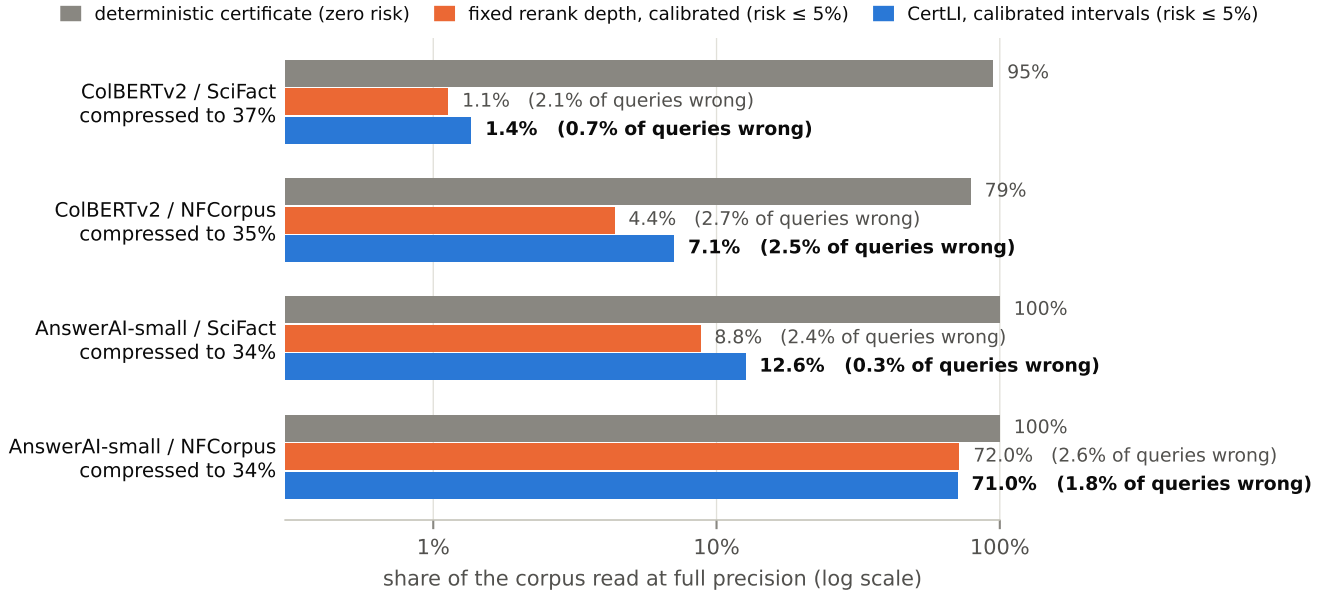


Figure 1: Share of the corpus scored at full precision to return the top-10, with the index compressed to about one third of its size. Grey: deterministic certificate (top-10 guaranteed exact). Blue: the same intervals shrunk by a factor chosen with Learn-then-Test, so that at most 5% of queries get a wrong top-10 (90% confidence). Orange: a fixed rerank depth calibrated the same way. In brackets: share of held-out queries whose top-10 differs from exact MaxSim, averaged over 200 random calibration/test splits.

1 Setup

Models and data. ColBERTv2 [6] (128-dimensional tokens) and AnswerAI-ColBERT-small-v1 (96 dimensions), encoded with PyLate using each model’s standard settings (queries expanded to 32 tokens, documents cut at 300 tokens). Corpora: BEIR SciFact (5,183 documents, 300 test queries) and NFCorpus (3,633 documents, 323 test queries). Ranking quality is nDCG@10 against exact MaxSim, $C(Q, D) = \sum_i \max_j \langle q_i, d_j \rangle$, computed over the whole corpus.

Compression. After subtracting the corpus mean μ , each document is stored in its own eigenbasis of at most 32 directions. Tokens that this basis represents poorly (residual above τ) are kept exactly; the rest are grouped into small clusters (radius at most ϵ), each stored with its own error radii. Both τ and ϵ are relative to the typical token norm, and sizes are reported as a share of the uncompressed fp16 index. For comparison, I certify two simpler codes with the same bounds: POOL, which clusters tokens without a subspace, and a PLAID-style residual code [8].

Cost. Every document gets an interval $[L_D, U_D]$ that is guaranteed to contain its exact score. Documents are rescored exactly in decreasing order of U_D until the next upper bound falls below the current 10th best score, which reads the fewest documents any correct method can read with these intervals [4]. I report this number as a share of the corpus. All runs used 2 CPU cores and no GPU.

2 Checking the pipeline

Before measuring certificates, I checked that the encoders, the evaluation and Lemma 1 behave as expected (Table 1).

Table 1: Pipeline checks: nDCG@10 of exact MaxSim (published value in brackets) and with IDF token weights [1] (change in brackets); Lemma 1, as the largest deviation of the shifted scores from the predicted constant offset and the share of queries with an identical top-100; nDCG@10 after subtracting each document’s own mean instead of the corpus mean.

Model	Data	nDCG@10 (published)	IDF-weighted nDCG@10	Lemma 1: max deviation	Same top-100	Per-document mean removed
ColBERTv2	SciFact	69.3 (69.3)	71.0 (+1.7)	3.2×10^{-6}	100%	68.0
ColBERTv2	NFCorpus	34.4 (33.8)	34.3 (0.0)	4.4×10^{-6}	100%	32.9
AnswerAI-small	SciFact	74.5 (74.8)	72.9 (−1.6)	3.8×10^{-6}	100%	37.7
AnswerAI-small	NFCorpus	37.0 (37.3)	36.8 (−0.2)	3.8×10^{-6}	100%	5.8

Exact MaxSim reproduces the published numbers within 0.6 points, and Lemma 1 holds up to float32 rounding: subtracting the corpus mean shifts all scores of a query by the same constant and leaves the top-100 unchanged for every query. Subtracting each document’s own mean is not safe and collapses AnswerAI-small (74.5 to 37.7), a first sign that this model’s tokens share a strong common direction. IDF token weights help ColBERTv2 on SciFact (+1.7) but hurt AnswerAI-small (−1.6), so useful token importance is model specific.

3 Results

3.1 The low-rank-plus-sparse structure is real

CertLI assumes that the tokens of a document lie close to a low-dimensional subspace, apart from a few tokens that do not fit. Table 2 tests this assumption.

Table 2: Geometry of document token sets. Token cosine: mean cosine between random tokens from different documents. Top-16 energy and effective rank of each document, before / after subtracting the corpus mean μ . $|\cos(v_1, \mu)|$: alignment of each document’s leading eigenvector with μ . Last two columns: how the residual left after a 16-dimensional projection depends on token IDF.

Model	Data	$\ \mu\ $	Token cosine	Top-16 energy %	Effective rank	$ \cos(v_1, \mu) $	Spearman (IDF, residual)	Residual, rarest / most common tokens
ColBERTv2	SciFact	0.48	0.23	80 / 74	16.6 / 28.5	0.76	0.45	2.0×
ColBERTv2	NFCorpus	0.49	0.24	81 / 75	15.3 / 26.4	0.75	0.48	2.0×
AnswerAI-small	SciFact	0.90	0.82	95 / 75	2.6 / 29.0	0.99	0.53	2.5×
AnswerAI-small	NFCorpus	0.92	0.84	96 / 76	2.4 / 28.2	0.99	0.55	2.5×

After the mean shift, 16 directions capture 74% to 76% of each document’s energy, and what remains is concentrated on rare tokens: the residual grows with IDF (Spearman 0.45 to 0.55) and is 2.0 to 2.5 times larger for the rarest tenth of tokens than for the most common tenth. This is why CertLI keeps rare tokens exactly. The table also shows that AnswerAI-small is strongly anisotropic: two random tokens from different documents have cosine 0.82 to 0.84, the corpus mean has norm 0.90 to 0.92 for unit-norm tokens, and each document’s leading eigenvector points along it.

3.2 Deterministic certificates score almost everything

With the structure confirmed, Table 3 measures what a deterministic certificate costs at different compression levels.

Table 3: Share of the corpus scored to certify the exact top-10, in % (index size as % of fp16 in brackets), for the low-rank-plus-sparse code at four settings and for POOL. Last column: bound violations over all configurations run.

Model	Data	low-rank + sparse, (τ, ϵ)				POOL $\epsilon = 0.45$	Violations / checks
		(0.3, 0.25)	(0.5, 0.25)	(0.5, 0.45)	(0.5, 0.45), global radii		
ColBERTv2	SciFact	39 (72)	78 (42)	95 (37)	100 (37)	78 (59)	0 / 21.8M
ColBERTv2	NFCorpus	28 (71)	58 (40)	79 (35)	99 (35)	58 (59)	0 / 8.2M
AnswerAI-small	SciFact	100 (65)	100 (42)	100 (34)	100 (34)	100 (53)	0 / 10.9M
AnswerAI-small	NFCorpus	100 (65)	100 (41)	100 (34)	100 (34)	100 (54)	0 / 9.4M

No bound was violated in any configuration, so the returned top-10 is always exact. The cost, however, is high. Even the largest code (72% of the index size) has to score 39% of SciFact for ColBERTv2; at about one third of the size, 79% to 95%; for AnswerAI-small, the whole corpus. The design choices still matter: per-cluster radii are needed (with one global radius the cost rises from 95% to 100%), and the subspace saves memory, since POOL needs 59% of the index size to reach the cost

that the low-rank code reaches at 40% to 42%. A 1-bit PLAID-style code (8% of the size, AnswerAI/NFCorpus) also has to score every document.

3.3 Why: the bounds pay for a worst case that rarely happens

Figure 2 and Table 4 show where this cost comes from, first at the level of single tokens and then at the level of whole documents.

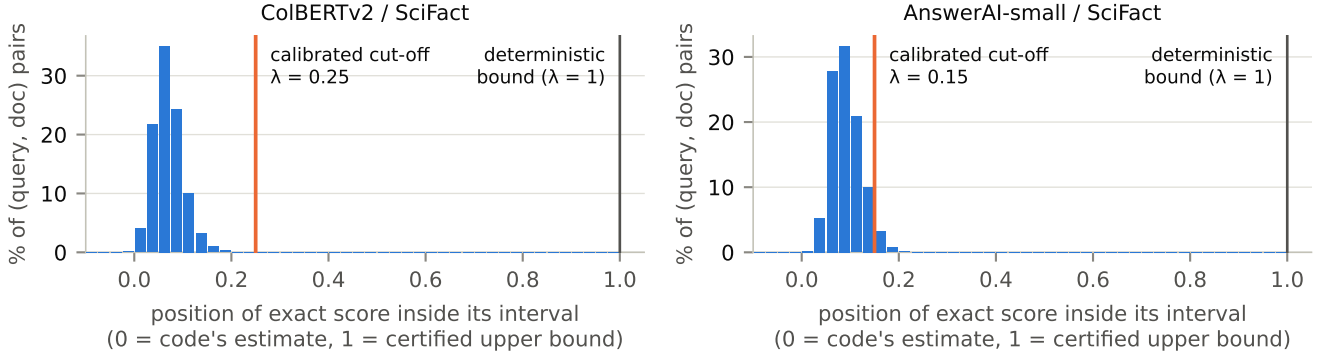


Figure 2: Where the exact score falls inside its interval, over all (query, document) pairs, with the index at about one third of its size. A deterministic certificate must allow for position 1; the factor chosen by Learn-then-Test sits just past where the exact scores actually end.

Table 4: Why certificates are loose. Tokens: RMS cosine between the query and the coding error (value for a random direction in brackets), and the actual error as a share of its bound (median / 99th percentile). Documents: median gap between the 10th and 11th exact scores, mean distance from the exact score to its upper bound, and how far the exact score sits inside its interval (99th percentile / maximum).

Model	Data	RMS cosine (random)	Error / bound	Top-10 gap	Mean distance to upper bound	Position p99 / max
ColBERTv2	SciFact	0.105 (0.094)	5% / 23%	0.078	11.3	16% / 49%
ColBERTv2	NFCorpus	0.101 (0.094)	5% / 22%	0.096	11.1	20% / 62%
AnswerAI-small	SciFact	0.131 (0.112)	5% / 27%	0.009	6.1	17% / 33%
AnswerAI-small	NFCorpus	0.137 (0.112)	5% / 26%	0.003	5.4	23% / 33%

Lemma 2 assumes the worst case, in which the coding error points exactly along the query. In practice the error direction is close to random: its RMS cosine with the query is only 1.07 to 1.22 times the value for a random direction, so the actual error is a median 5% of its bound. Figure 2 shows the consequence for whole documents: the exact score sits a median 5% to 14% of the way from the code’s estimate to the upper bound, and never beyond 62%. At the same time, the top-10 is decided by very small margins. The median gap between the 10th and 11th exact scores is 0.08 to 0.10 for ColBERTv2 and 0.003 to 0.009 for AnswerAI-small, while the upper bounds sit 5 to 11 above the exact scores on average. A worst-case bound cannot separate scores this close, but a bound sized for the typical case could.

3.4 Calibrated certificates

This suggests shrinking each interval towards the code’s estimate and choosing the shrink factor λ from data. Table 5 does this with Learn-then-Test, which guarantees that at most 5% of queries get a wrong top-10, with 90% confidence, and compares the result with two baselines.

Table 5: Calibrated certificates (Learn-then-Test, $\alpha = 0.05$, $\delta = 0.1$, 200 random 50/50 calibration/test splits). Share of the corpus scored, in %, with the share of held-out queries that get a wrong top-10 in brackets. $\hat{\lambda}$: selected shrink factor ($\lambda = 1$ is the deterministic certificate). Oracle: a perfect per-query stopping rule.

Model	Data	Index size	Deterministic	Calibrated CertLI	$\hat{\lambda}$	Calibrated fixed depth	Oracle
ColBERTv2	SciFact	37%	94.8	1.4 (0.7)	0.25	1.1 (2.1)	0.4
ColBERTv2	NFCorpus	35%	78.9	7.1 (2.5)	0.4	4.4 (2.7)	0.7
AnswerAI-small	SciFact	34%	100.0	12.6 (0.3)	0.15	8.8 (2.4)	1.3
AnswerAI-small	NFCorpus	34%	100.0	71.0 (1.8)	0.225	72.0 (2.6)	14.4

The selected $\hat{\lambda}$ is 0.15 to 0.40, in line with where the exact scores actually fall. On ColBERTv2/SciFact, CertLI now scores 70 of 5,183 documents per query on average (1.4%), and only 0.7% of held-out queries get a wrong top-10. Across all four settings the observed error stays between 0.3% and 2.5%. Two points need to be stated clearly. First, a fixed rerank depth calibrated in the same way is just as cheap or cheaper, with a higher error, so for now the gain comes from calibration rather than from adapting to each query. Second, there is real room for adaptivity: a perfect per-query stopping rule would score

only 0.4% to 14.4% of the corpus, 3 to 7 times less than the best fixed depth, and its depth correlates with how crowded the scores near the 10th are (Spearman 0.36 to 0.69), which suggests it can be learned. AnswerAI/NFCorpus remains expensive (71%) because its score gaps are the smallest.

3.5 Side result: the shared direction and EigenLI

The shared direction in AnswerAI-small also matters for spectral methods such as EigenLI [2]. Table 6 compares, at equal storage, my re-implementation of EigenLI’s projection score with and without removing the corpus mean, together with Ward token pooling.

Table 6: SciFact, k vectors per document (equal storage in every column): nDCG@10, with overlap with the exact top-10 in % in brackets; exact MaxSim next to the model name. EigenLI is my re-implementation of the projection score $\sum_i \|\Pi_D q_i\|^2$, with Π_D the top- k eigenspace of the document’s tokens [2]. Ward: hierarchical token pooling [7].

Model (exact)	k	EigenLI	Mean removed, documents only	Mean removed, documents and queries	Ward pooling
ColBERTv2 (69.3)	16	63.6 (56)	63.5 (56)	60.3 (49)	56.7 (44)
ColBERTv2 (69.3)	32	67.7 (64)	67.9 (63)	66.6 (59)	61.0 (55)
AnswerAI-small (74.5)	16	68.0 (53)	1.6 (2)	68.2 (52)	60.0 (41)
AnswerAI-small (74.5)	32	67.4 (55)	0.3 (1)	69.6 (56)	63.0 (43)

Removing the mean from documents only destroys the ranking for AnswerAI-small (68.0 to 1.6 at $k = 16$): queries keep their large shared component, which then falls outside every document’s subspace and turns into noise. Removing it from both documents and queries is neutral at $k = 16$ and helps at $k = 32$ (67.4 to 69.6), while for the less anisotropic ColBERTv2 it costs 1.1 to 3.3 points. So subtracting the mean on one side is free for MaxSim (Lemma 1) but not for quadratic scores, and the right treatment of the shared direction depends on the model. In every setting, EigenLI beats Ward pooling by 4.4 to 8.0 points.

4 Takeaways and next steps

What worked. Both lemmas hold and are cheap to apply. The low-rank-plus-sparse structure is real and saves about 30% of memory over plain clustering at the same certification cost. Calibrated certificates control the error while scoring 1.4% to 12.6% of the corpus in three of the four settings.

What did not. Deterministic certificates still score most of the corpus at useful compression, because worst-case bounds ignore that the coding error is nearly random in direction and that top-10 margins are tiny. Adapting the stopping point to each query does not yet beat a calibrated fixed depth.

Next steps. (1) Apply a random rotation or dither [5] before compression, so that the near-isotropy observed here becomes a guarantee; the bound can then rely on concentration instead of the worst case and give a high-probability certificate without calibration data. (2) Learn the stopping rule from score margins and density, aiming at the 3 to 7 times headroom shown by the oracle. (3) Scale to MS MARCO and LoTTE with PLAID candidate generation on GPU, certifying PLAID’s own compressed codes.

Scope. Small corpora (3.6k and 5.2k documents), CPU only, exact scoring of the full corpus as ground truth, and 150 to 162 calibration queries per split. EigenLI is my re-implementation, not the authors’ code.

References

- [1] Archish S, A. Garg, K. Shiragur, N. Kayal. Incorporating token importance in multi-vector retrieval. AAAI 2026. [2] Archish S, S. Basu, A. Garg, R. Krishnaswamy, K. Shiragur. EigenLI: spectral approximations to late interaction. arXiv:2609.07561, 2026. [3] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, L. Lei. Learn then Test: calibrating predictive algorithms to achieve risk control. arXiv:2110.01052, 2021. [4] T. Seidl, H.-P. Kriegel. Optimal multi-step k -nearest neighbor search. SIGMOD 1998. [5] L. Schuchman. Dither signals and their effect on quantization noise. IEEE Trans. Communication Technology, 1964. [6] K. Santhanam, O. Khattab, J. Saad-Falcon, C. Potts, M. Zaharia. ColBERTv2: effective and efficient retrieval via lightweight late interaction. NAACL 2022. [7] B. Clavié, A. Chaffin, G. Adams. Reducing the footprint of multi-vector retrieval with minimal performance impact via token pooling. arXiv:2409.14683, 2024. [8] K. Santhanam, O. Khattab, C. Potts, M. Zaharia. PLAID: an efficient engine for late interaction retrieval. CIKM 2022.